Script Relationships

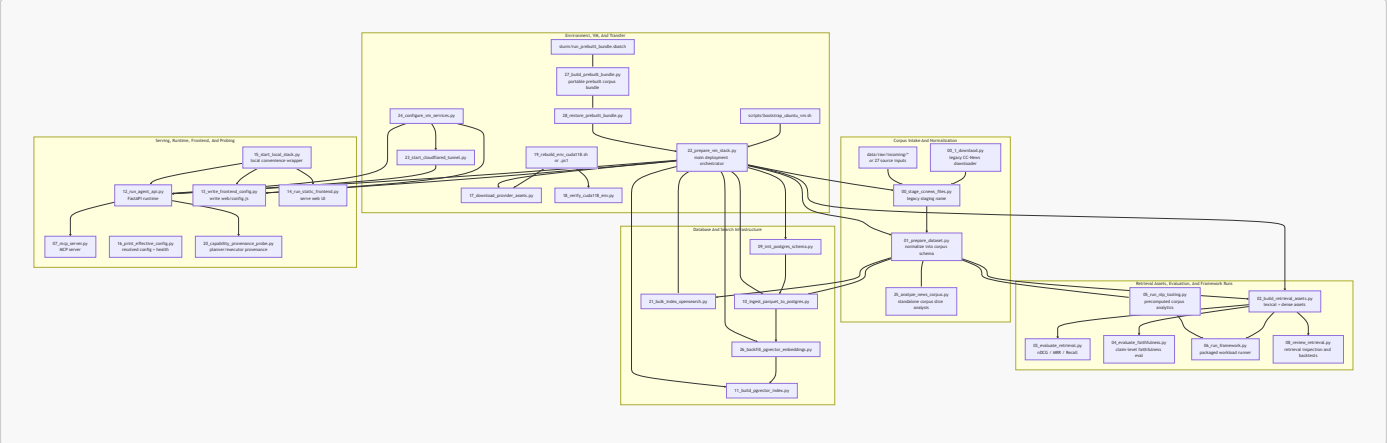
This is the script-centric map of the repo: which scripts work together, what they produce, and where they sit in the larger corpus-agent workflow.

CorpusAgent2 should be understood as a corpus agent over a normalized corpus, not as a news-only system. Some script names still contain `ccnews` or `news`; that is legacy naming from the original example corpus and does not define the architecture boundary.

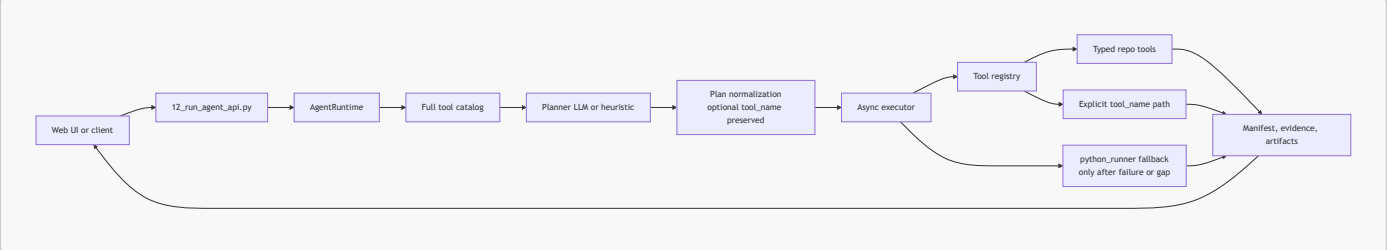
How To Read The Numbering

- `00` to `05`: corpus intake, normalization, retrieval assets, evaluation, and precomputed NLP artifacts
- `06` to `08`: packaged experiments, MCP exposure, and retrieval inspection
- `09` to `11`, `21`, `26`: Postgres, pgvector, and OpenSearch ingestion/indexing
- `12` to `16`: API, frontend config, local serving, and config inspection
- `17` to `19`: provider assets and CUDA-capable environment setup
- `20`: planner and executor provenance probing
- `22` to `24`: VM bootstrap, tunnels, and persistent service setup
- `25`: standalone corpus slice analysis
- `27` to `28`: cluster-side portable bundle build and VM-side restore

Script Orchestration Workflow



Runtime Decision Workflow



What Each Group Is For

- Intake scripts turn raw files or HF rows into the common corpus schema.
- Asset scripts build the retrieval layer and optional precomputed analytics the runtime can reuse.
- Storage scripts move the normalized corpus into Postgres, pgvector, and OpenSearch for larger deployments.
- Serving scripts expose the agent through FastAPI, the static UI, or MCP.
- Deployment scripts make the same repo runnable on a VM or on a GPU cluster with a prebuilt transfer bundle.

Key Artifacts And Directories

Path	Produced by	Used by
`data/raw/incoming/`	manual input, `00_1_downlaod.py`, `27_build_prebuilt_bundle.py`	`00_stage_ccnews_files.py`
`data/raw/ccnews_staged/`	`00_stage_ccnews_files.py`	`01_prepare_dataset.py`
`data/processed/documents.parquet`	`01_prepare_dataset.py`	`02_build_retrieval_assets.py`, `10_ingest_parquet_to_postgres.py`, `21_bulk_index_opensearch.py`, `25_analyze_news_corpus.py`
`data/indices/lexical/*`	`02_build_retrieval_assets.py`	retrieval runtime, `03_evaluate_retrieval.py`, `08_review_retrieval.py`

Path	Produced by	Used by
`data/indices/dense/*`	`02_build_retrieval_assets.py`	retrieval runtime, `03_evaluate_retrieval.py`, `10_ingest_parquet_to_postgres.py`, `26_backfill_pgvector_embeddings.py`
`data/indices/doc_metadata.parquet`	`02_build_retrieval_assets.py`	runtime, evaluation, and review
`outputs/nlp_tools/*`	`05_run_nlp_tooling.py`	runtime reuse, `06_run_framework.py`, bundle/VM flow
`outputs/retrieval_eval/*`	`03_evaluate_retrieval.py`	evaluation review/reporting
`outputs/faithfulness_eval/*`	`04_evaluate_faithfulness.py`	evaluation review/reporting
`outputs/postgres/*`	`09`, `10`, `11`, `26`	deployment diagnostics
`outputs/opensearch/*`	`21_bulk_index_opensearch.py`	deployment diagnostics
`outputs/agent_runtime/*`	`12_run_agent_api.py` runtime queries	UI run inspection and artifacts
`outputs/prebuilt/*`	`27_build_prebuilt_bundle.py`	`28_restore_prebuilt_bundle.py`, VM transfer

Entry Points By Goal

Goal	Normal entry point	What it wraps
Build a local corpus from raw files	`00 -> 01 -> 02`	staging, normalization, retrieval assets
Add precomputed corpus analytics	`05_run_nlp_tooling.py`	sentiment/entity/topic/time-slice artifacts
Evaluate retrieval quality	`03_evaluate_retrieval.py`	retrieval metrics over gold queries
Evaluate answer faithfulness	`04_evaluate_faithfulness.py`	NLI-style support checks over gold claims
Run API + frontend locally	`15_start_local_stack.py`	`13_write_frontend_config.py`, `12_run_agent_api.py`, `14_run_static_frontend.py`
Run backend only	`12_run_agent_api.py`	FastAPI runtime only
Bootstrap a VM end-to-end	`22_prepare_vm_stack.py`	environment, assets, Docker services, indexes, frontend config, optional API start
Preprocess on a GPU cluster and move to VM	`slurm/run_prebuilt_bundle.sbatch` -> `27_build_prebuilt_bundle.py`	cluster preprocessing + portable zip
Restore preprocessed corpus artifacts on VM	`28_restore_prebuilt_bundle.py`	unzip into repo so VM bootstrap can skip slow prep

Script Reference

Script	Main job	Main inputs	Main outputs	Works v
`00_1_downlaod.py`	Download the original example `cc_news` dataset directly into raw incoming	hard-coded HF dataset id	`data/raw/incoming/cc_news.jsonl.gz`	legacy q
`00_stage_ccnews_files.py`	Copy raw corpus files into a clean staged area and write a summary manifest	`data/raw/incoming/`	`data/raw/ccnews_staged/`, stage summary JSON	`01_prep` VM boo
`01_prepare_dataset.py`	Normalize staged rows into the repo corpus schema	staged raw files	`data/processed/documents.parquet`, summary JSON	`02`, `10`, 2
`02_build_retrieval_assets.py`	Build lexical and optional dense retrieval assets plus metadata parquet	`data/processed/documents.parquet`	`data/indices/lexical/*`, `data/indices/dense/*`, `data/indices/doc_metadata.parquet`	runtime,
`03_evaluate_retrieval.py`	Score retrieval quality with `nDCG`, `MRR`, `Recall`, bootstrap CI, paired tests	retrieval assets + gold query file	`outputs/retrieval_eval/*`	Slurm ev
`04_evaluate_faithfulness.py`	Score claim support / faithfulness against gold claims	claim file + corpus metadata	`outputs/faithfulness_eval/*`	Slurm ev

Script	Main job	Main inputs	Main outputs	Works v
`05_run_nlp_tooling.py`	Precompute corpus-level NLP artifacts such as sentiment/entity/topic time series	processed documents + granularity config	`outputs/nlp_tools/*`	runtime
`06_run_framework.py`	Run the framework workload file over the prepared corpus	workload file + corpus assets	framework output directory	usually p
`07_mcp_server.py`	Expose the runtime as an MCP server over stdio	repo runtime/config	long-running MCP process	editor/to
`08_review_retrieval.py`	Inspect retrieval payloads or backtest them against gold queries	retrieval output JSON/JSONL, gold queries, assets	`outputs/retrieval_audit/*`	compler
`09_init_postgres_schema.py`	Create pgvector extension, corpus table, and helper indexes	Postgres DSN env	Postgres init summary	`10`, `22`
`10_ingest_parquet_to_postgres.py`	Upsert normalized corpus documents into Postgres	`data/processed/documents.parquet`, optional dense assets	Postgres ingest summary	`09`, `26`, 2
`11_build_pgvector_index.py`	Build ANN indexes in Postgres for dense retrieval	populated Postgres corpus table	Postgres index summary	`26`, `22`
`12_run_agent_api.py`	Start the FastAPI backend and corpus-agent runtime	config, retrieval assets, optional Postgres/OpenSearch	live API server	`15`, `22`, 2
`13_write_frontend_config.py`	Write runtime config into `web/config.js`	app config + env overrides	`web/config.js`	`14`, `15`, 2
`14_run_static_frontend.py`	Serve the static `web/` frontend locally	`web/` and generated config	live frontend server	`15`
`15_start_local_stack.py`	Convenience launcher for backend + frontend together	repo config	starts `12`, writes config with `13`, starts `14`	local dev
`16_print_effective_config.py`	Print resolved config and retrieval health for debugging	config + env	stdout JSON	preflight
`17_download_provider_assets.py`	Download external NLP assets for spaCy, stanza, NLTK, TextBlob	active Python env	provider model/data files	`19`, `22`
`18_verify_cuda118_env.py`	Check whether the CUDA-capable environment is usable	active Python env	stdout GPU/module report	`19`
`19_rebuild_env_cuda118.sh` / `.ps1`	Rebuild the virtualenv for CUDA 11.8 and verify it	repo + `uv`	fresh `.venv`, requirements, provider assets	`17`, `18`
`20_capability_provenance_probe.py`	Run representative queries and print planner/tool provenance	local runtime + example queries	stdout provenance summaries	debuggi
`21_bulk_index_opensearch.py`	Create the OpenSearch index and bulk-load the normalized corpus	processed documents + OpenSearch config	OpenSearch index summary	`22`
`22_prepare_vm_stack.py`	Main bootstrap/orchestration script for VM deployment	env, repo, optional raw data or restored bundle	full stack setup, summaries, optional API start	wraps m
`23_start_cloudflared_tunnel.py`	Start a quick Cloudflare tunnel for the local	running API	public tunnel URL in stdout	demos/t

Script	Main job	Main inputs	Main outputs	Works v
	backend			
`24_configure_vm_services.py`	Install reboot-safe `systemd` services for API and optional tunnel	VM host with sudo/systemd	service files + enabled services	`13`, `12`, `2`
`25_analyze_news_corpus.py`	Run a standalone deep-dive analysis for a selected corpus slice	processed corpus parquet	analysis JSON/parquet outputs in custom dir	separate
`26_backfill_pgvector_embeddings.py`	Fill Postgres `dense_embedding` values from local dense assets or on-the-fly encoding	Postgres rows + optional dense assets	Postgres backfill summary	`10`, `11`, `2`
`27_build_prebuilt_bundle.py`	Build a VM-ready portable bundle from a HF corpus or local files	HF dataset or source file(s)	`outputs/prebuilt/*.zip`, bundle manifest	`slurm/r` `28`
`28_restore_prebuilt_bundle.py`	Restore the portable bundle into the repo tree on the VM	bundle zip from `27`	restored `data/processed`, `data/indices`, optional `outputs/nlp_tools`	`22`
`bootstrap_ubuntu_vm.sh`	Thin Ubuntu wrapper around VM bootstrap	repo path / shell env	calls `22_prepare_vm_stack.py`	quick VM

Slurm Wrappers

Slurm script	What it runs
`slurm/run_prepare.sbatch`	`00_stage_ccnews_files.py` -> `01_prepare_dataset.py`
`slurm/run_build_assets.sbatch`	`02_build_retrieval_assets.py`
`slurm/run_evaluation.sbatch`	`03_evaluate_retrieval.py` -> `04_evaluate_faithfulness.py`
`slurm/run_framework.sbatch`	`05_run_nlp_tooling.py` -> `06_run_framework.py`
`slurm/run_prebuilt_bundle.sbatch`	`27_build_prebuilt_bundle.py --clean-existing ...`

Most Common Real Workflows

1. Local corpus plus local UI

```
00_stage_ccnews_files.py -> 01_prepare_dataset.py -> 02_build_retrieval_assets.py -> 15_start_local_stack.py
```

2. Local corpus plus precomputed analytics

```
00_stage_ccnews_files.py -> 01_prepare_dataset.py -> 02_build_retrieval_assets.py -> 05_run_nlp_tooling.py ->
12_run_agent_api.py
```

3. Postgres plus pgvector hybrid retrieval

```
09_init_postgres_schema.py -> 10_ingest_parquet_to_postgres.py -> 26_backfill_pgvector_embeddings.py ->
11_build_pgvector_index.py -> 12_run_agent_api.py
```

4. OpenSearch plus VM deployment

```
22_prepare_vm_stack.py
```

5. GPU cluster preprocessing plus VM serving

```
slurm/run_prebuilt_bundle.sbatch -> 27_build_prebuilt_bundle.py -> download zip -> 28_restore_prebuilt_bundle.py ->
22_prepare_vm_stack.py
```

Practical Notes

- `01_prepare_dataset.py` is the main schema boundary. If your corpus can be normalized there, the rest of the stack treats it as generic corpus data.
- `22_prepare_vm_stack.py` is the densest script because it is an orchestrator that wraps many other scripts.
- `15_start_local_stack.py` is only a convenience wrapper; the real serving logic lives in `12`, `13`, and `14`.
- `27` and `28` are the cleanest path when preprocessing is too slow on the VM.
- `00_1_downlaod.py`, `00_stage_ccnews_files.py`, and `25_analyze_news_corpus.py` keep their historical names, but the broader pipeline is not restricted to news as a domain.